Isaac Newton – Founder of classical physics 18a



Isaac Newton was born into a farming family in Woolsthorpe, England, in 1643. He went to the Free Grammar School in nearby Grantham. His school reports described him as idle and inattentive. Isaac mostly kept his own company and was known as a sober, silent, thinking lad. When he did associate with others, it was nearly always with girls. Isaac is remembered for his strange inventions and extraordinary inclination for mechanical works. Among these was a windmill powered by a treadmill run by a mouse.

Later, his mother made Isaac leave school in order to help run the family farm. But with support from his uncle, Isaac continued his studies at Trinity College, Cambridge, in 1661. Newton's aim at Cambridge was a law degree, but soon he became involved with other topics like algebra, optics and astronomy. Newton did a lot of studying of other scientists and ancient philosophers.

To test a question about colours, he stared at the sun with one eye until all the colours changed. As a result, he had to shut himself up in a dark room for several days before he could get rid of the spots floating before his eyes.

Despite some evidence that his progress had not been particularly good, Newton received his bachelor's degree in 1665. His scientific genius suddenly emerged when the plague closed the university in the summer of 1665 and he had to return home to Woolsthorpe. There, in a period of less than two years, he began revolutionary advances in mathematics, optics, physics and astronomy. In 1666 Newton had early versions of his three laws of motion. Newton had also reached the conclusion that white light is not a single entity, as most scientists since Aristotle had believed. When Newton passed a thin beam of sunlight through a triangular glass prism, the spectrum of the rainbow colours formed. However, Newton did not publish any of his work then.

When the University of Cambridge reopened in 1667 after the plague, Newton was elected to a major fellowship within a year.

Newton then became interested in alchemy. He laboured day and night in his chemical laboratory trying to produce gold from other metals. And then he immersed himself in mathematical and mystical calculations. In the late 1670s, theological studies occupied most of his time.

In 1686 Newton presented his single greatest work, the *Philosophiae Naturalis Principia Mathematica* ('Mathematical Principles of Natural Philosophy', simply known as the 'Principia').

In it, Newton revealed his laws of motion and the law of universal gravitation. The first law of motion was about inertia: a body wants to keep doing what it is doing. A moving body wants to continue moving in a straight line, while a stationary body wants to remain stationary.

The second law ($a = F / m$) said that the acceleration of an object depends on the force (stronger force means more acceleration) and its mass (bigger mass means less acceleration).

The third law stated every action force has an equal but opposite reaction force. Newton further demonstrated that all heavenly bodies mutually attract one another. Further generalisation led Newton to the law of universal gravitation. The astronomer Edmund Halley (of Halley's comet fame) paid for the publication of the Principia. The Principia redirected Newton's intellectual life away from theology and alchemy, back into science.

After suffering a second nervous breakdown in 1693, Newton retired from research and decided to leave Cambridge for a government position in London. He became the Warden of the Royal Mint in 1696 and Master in 1699. In 1703 he was elected president of the Royal Society and was re-elected each year until his death.

Newton was knighted by Queen Anne in 1705; the first scientist to be honoured in this way. His last decades were passed in revising his major works, polishing his studies of ancient history, and defending himself against critics, as well as carrying out his official duties. Newton was modest, diffident and a man of simple tastes. He was angered by criticism or opposition, and he harboured resentment; he was harsh towards enemies but generous to friends. He never married and lived modestly.

In 1709, Newton began work on a second edition of Principia and he also published a second edition of Opticks; however after he moved to London, he did nothing but reshuffle his old ideas. As his health began to deteriorate, he began to distribute his wealth amongst his family. He died in 1727 after a series of debilitating illnesses.

© Chris Greef, Suada Bilali and Blake Education 2006

Isaac Newton – Founder of classical physics 18b



Complete these tasks in your science book.

1 Make a timeline starting at 1640 and finishing in 1760. Mark and number every other decade above the line; mark and label Newton's age in increments of ten years below the line. Then indicate on the timeline the major events in Isaac Newton's life.

Reading on the lines ...

2 Why did Newton leave school?

3 What were Newton's interests at school?

4 School and university were not free in Newton's time. Who paid for his education?

5 What subject did Newton start studying at university?

6 Why did Newton return home after his university studies?

7 How long was there between Newton's graduation and his election to a major fellowship?

8 What is the short title for Newton's major book?

9 What is inertia?

10 What made Edmond Halley famous?

Reading between the lines ...

11 How old was Newton when he died?

12 How old was Newton when he started university?

13 Why would Newton have stared at the sun with only one eye?

14 What did an alchemist try to do?

15 How many years after conception of his ideas did Newton publish them?

16 Forces usually occur in pairs. What are their names?

17 What is the name given to the attraction between two masses?

18 Why did Newton leave university life?

19 How did Newton earn his money in London?

20 What exciting event happened to Newton in 1705?

21 Newton's scientifically creative period occurred during his two years in the country after he had graduated from university. Back in London, what scientific work did he do?

Reading beyond the lines ...

22 Why would Newton's mother have made him leave school?

23 What is wrong with staring at the sun for any length of time?

24 What is the scientific term for the bending of light in a prism?

25 What are the seven colours of the rainbow spectrum?

26 Why didn't Newton publish his ideas straight after conception?

27 What do the letters in $\mathbf{a = F/m}$ stand for?

28 How do you calculate the force needed to make an object move?

29 Why is Halley's comet named so?

30 Why would Newton have given his wealth to his relations?

31 Newton said that he stood 'on the shoulders of giants' when he developed his laws. What did he mean and who were these giants?

© Chris Greef, Suada Bilali and Blake Education 2006